

Intraspecific sexual isolation in *Drosophila*

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Intraspecific sexual isolation was examined among wild strains of *Drosophila malerkotliana*, *D. parabipectinata* and *D. pseudoananassae* by multiple-choice method in Elens-Wattiaux mating chamber. In *D. pseudoananassae*, mating between two strains tested was random and isolation estimate was close to one. In one out of 6 crosses, involving geographic strains of *D. malerkotliana*, there was significant deviation from randomness and isolation estimate remained low which shows non-random (preferential or positive assortative) mating. In *D. parabipectinata*, the deviation from randomness was statistically significant due to higher number of homogamic matings in three crosses involving wild strains derived from geographically distant places and isolation estimate remained low in these crosses. The results provide evidence for incipient sexual isolation within *D. malerkotliana* and *D. parabipectinata* as a result of genetic divergence.

Dobzhansky¹ coined the term "isolating mechanisms" for genetically conditioned barriers which prevent gene flow between Mendelian populations. Sexual (ethological) isolation constitutes the most important class among different means of reproductive isolation in animal species and thus it plays an important role in speciation. Due to behavioural incompatibility the opposite sexes of different populations fail to mate which does not allow one gene pool dispersing freely into a foreign gene pool. Thus ethological isolation is a premating barrier to gene exchange. The phenomenon of behavioural isolation has been extensively studied in the genus *Drosophila* and has been found to be widespread²⁻⁴. The results obtained in different species have clearly shown that complete sexual isolation exists between some species and incomplete between others. Allopatric populations of the same species may be differentiated into races due to modification of gene frequencies by the action of natural selection and/or genetic drift¹. Races may attain the status of species if they develop some kind of reproductive isolation. In a number of *Drosophila* species, incipient sexual isolation has been found to occur between geographic strains of the same species²⁻⁴.

Incomplete sexual isolation has been observed among four species of the *bipectinata* complex, *D. bipectinata*, *D. parabipectinata*, *D. malerkotliana* and *D. pseudoananassae*^{5,6,7}. Bock⁷ did not find any evidence of incipient sexual isolation among geographic strains of these species. However, Singh & Chatterjee (unpublished) found evidence for incipient sexual isolation within *D. bipectinata* when

six geographic strains were tested. During the present study, we examined intraspecific sexual isolation in *D. malerkotliana*, *D. parabipectinata* and *D. pseudoananassae* by using different geographic strains and the results are reported in this communication.

Materials and Methods

To test intraspecific sexual isolation, the following strains of three species were used:

D. malerkotliana—(1) BHU Campus, Varanasi; (2) Unchawa Lodge near U.P. College, Varanasi; (3) Mysore, Karnataka; and (4) Baripada, Orissa.

D. parabipectinata—(1) Mysore, Karnataka; (2) Cambodia; (3) Indonesia; and (4) Mauritius (Isle of Redrigues)

D. pseudoananassae—(1) Thailand (Chiang Mai); and (2) Australia (North Queensland).

All these strains were initiated from flies collected from natural populations and had spent varying number of generations in the laboratory before they were tested. To test sexual isolation among different strains of the same species, multiple-choice method was used and mating was observed in Elens-Wattiaux mating chamber. Virgin females and males were collected and aged for 7 days in small batches. In all the experiments 15 flies of each sex were used and 5 trials were run for each experimental set. Females (15) of each of two strains were introduced in the mating chamber with 15 males of each of two strains and were observed for 60 min. When a pair commenced mating, it was aspirated out and the types of individuals mated were recorded. All the crosses were run at approximately 24°C and during

0700-1100 hrs. To identify the females and males of different strains in each cross, the flies of one stock were marked with a small spot of fast drying nail polish on the scutellum. Before making the cross, the effect of marking was tested and no effect of marking was found either on the performance of flies or outcome of sexual isolation tests. The experiment was repeated by marking the flies of other strain. The results of both experiments were pooled.

To measure the degree of sexual isolation among different strains, isolation estimate was calculated using the formula of Merrell⁸:

$$\text{Isolation estimate} = \frac{\text{number of heterogamic matings}}{\text{number of homogamic matings}}$$

The isolation estimate ranges from zero to infinity. If isolation estimate is one, there is no sexual isolation between the strains tested. If it is zero, then isolation is complete. The χ^2 values were calculated to measure the differences between homogamic and heterogamic matings under the assumption of random mating. Any significant deviation from randomness would indicate non-random (preferential or assortative) mating.

Results

Results of mutiple-choice experiments involving four geographic strains of *D. malerkotliana* are presented in Table 1. In 5 crosses, isolation estimate was close to one and deviation from randomness was not significant which indicates random mating and no isolation. However, in one cross involving Baripada and BHU strains, the number of

homogamic matings was more than heterogamic ones and deviation from randomness was statistically significant ($P < 0.05$). The value of isolation estimate was also very low in this cross. Thus these two strains of *D. malerkotliana* show incipient sexual isolation.

These was no difference between homogamic and heterogamic matings in the cross involving two strains of *D. pseudoananassae* originating from Australia and Thailand and isolation estimate remained close to one which indicates no evidence for sexual isolation within *D. pseudoananassae* (Table 2). In *D. parabipectinata*, six crosses were made among four strains derived from distant localities (Table 2). In 3 crosses i.e. Mauritius $\times$ Cambodia; Mauritius $\times$ Mysore and Cambodia $\times$ Indonesia, isolation estimate was close to one and there was no significant deviation from randomness which indicates random mating. On the other hand, homogamic matings were more frequent than heterogamic ones in three crosses viz. Mauritius $\times$ Indonesia, Cambodia $\times$ Mysore and Mysore $\times$ Indonesia and differences were statistically significant ($P < 0.05$). Further there was considerable decrease in the value of isolation estimate in these crosses due to less number of heterogamic matings. In these crosses, the females of one strain preferred its own males and discriminated against alien males. This indicates the existence of incipient sexual isolation within *D. parabipectinata*.

Discussion

Drosophila bipectinata species complex belongs to *ananassae* subgroup of *melanogaster* species group.

Table 1—Numbers of matings in 60 min in mating chamber between different strains of *Drosophila malerkotliana* in multiple-choice experiments

Strains	Numbers of homogamic and heterogamic matings				Isolation Estimate	χ^2
	Homo.	Het.	Homo.	Het.		
BHU Campus $\times$ Unchawa Lodge	BHU♀ $\times$ BHU♂ 38	BHU♀ $\times$ UL♂ 43	UL♀ $\times$ UL♂ 38	UL♀ $\times$ BHU♂ 32	0.99	0.01
BHU Campus $\times$ Mysore	BHU♀ $\times$ BHU♂ 38	BHU♀ $\times$ My♂ 38	My♂ 40	My♀ $\times$ BHU♂ 33	0.91	0.33
BHU Campus $\times$ Baripada	BHU♀ $\times$ BHU♂ 42	BHU♀ $\times$ Br♂ 29	Br♀ $\times$ Br♂ 45	Br♀ $\times$ BHU♂ 29	0.67	5.80*
Unchawa Lodge $\times$ Mysore	UL♀ $\times$ UL♂ 40	UL♀ $\times$ My♂ 35	My♀ $\times$ My♂ 39	My♀ $\times$ UL♂ 37	0.91	0.32
Unchawa Lodge $\times$ Baripada	UL♀ $\times$ UL♂ 33	UL♀ $\times$ Br♂ 37	Br♀ $\times$ Br♂ 45	Br♀ $\times$ UL♂ 35	0.92	0.24
Mysore $\times$ Baripada	My♀ $\times$ My♂ 38	My♀ $\times$ Br♂ 39	Br♀ $\times$ Br♂ 35	Br♀ $\times$ My♂ 38	1.05	0.11

* Significant (df = 1)

Table 2—Numbers of matings in 60 min in mating chamber between different strains of *Drosophila pseudoananassae* and of *D. parabipectinata* in multiple-choice experiments

Species and strains	Numbers of homogamic and heterogamic matings				Isolation Estimate	χ^2
	Homo.	Het.	Homo.	Het.		
<i>D. pseudoananassae</i>						
Australia $\times$ Thailand	Aus♀ $\times$ Aus♂ 37	Aus♀ $\times$ Thai♂ 34	Thai♀ $\times$ Thai♂ 40	Thai♀ $\times$ Aus♂ 37	0.92	0.24
<i>D. parabipectinata</i>						
Mauritius $\times$ Cambodia	Mau♀ $\times$ Mau♂ 45	Mau♀ $\times$ Cam♂ 31	Cam♀ $\times$ Cam♂ 35	Cam♀ $\times$ Mau♂ 36	0.84	1.14
Mauritius $\times$ Mysore	Mau♀ $\times$ Mau♂ 46	Mau♀ $\times$ My♂ 32	My♀ $\times$ My♂ 33	My♀ $\times$ Mau♂ 39	0.90	0.43
Mauritius $\times$ Indonesia	Mau♀ $\times$ Mau♂ 35	Mau♀ $\times$ Ind♂ 28	Ind♀ $\times$ Ind♂ 61	Ind♀ $\times$ Mau♂ 25	0.55	12.41*
Cambodia $\times$ Mysore	Cam♀ $\times$ Cam♂ 48	Cam♀ $\times$ My♂ 32	My♀ $\times$ My♂ 40	My♀ $\times$ Cam♂ 29	0.69	4.89*
Cambodia $\times$ Indonesia	Cam♀ $\times$ Cam♂ 38	Cam♀ $\times$ Ind♂ 33	Ind♀ $\times$ Ind♂ 40	Ind♀ $\times$ Cam♂ 41	0.95	0.11
Mysore $\times$ Indonesia	My♀ $\times$ My♂ 41	My♀ $\times$ Ind♂ 26	Ind♀ $\times$ Ind♂ 60	Ind♀ $\times$ My♂ 24	0.50	17.23*

* Significant (df = 1)

Four closely related members of this complex are distributed in Australian and/or Oriental biogeographic zones. A close phylogenetic relationship among these species has been demonstrated on the basis of chromosomal studies⁹ and degree of interspecific hybridization^{5,7}. Bock⁷ also studied intraspecific sexual isolation in *D. bipectinata*, *D. parabipectinata*, *D. malerkotliana* and *D. pseudoananassae* by using geographic strains from Borneo, Thailand and Cairns. In all the crosses (except one), the differences between homogamic and heterogamic matings were not significant and thus no evidence of incipient sexual isolation was found among any of the strains of 4 species tested. In one cross (*D. pseudoananassae*), difference was significant but isolation index was negative due to higher number of heterogamic matings⁷. Singh & Chatterjee (unpublished) also studied intraspecific sexual isolation within *D. bipectinata* by employing 6 geographic strains originating from widely distant localities. The multiple-choice method was used and mating was directly observed in mating chamber. The results indicated that in all the crosses homogamic matings were more frequent than heterogamic ones and deviation from randomness was significant in six crosses. Based on this, it was suggested that incipient sexual isolation exists within *D. bipectinata* which might have originated as a byproduct of genetic divergence (Singh and Chatterjee, unpublished).

During the present study, intraspecific sexual isolation was tested within *D. parabipectinata*, *D. malerkotliana* and *D. pseudoananassae* by using different geographic strains of these species. In *D. pseudoananassae*, two strains tested did not show sexual isolation as mating between them was random and isolation estimate was nearly one. In *D. malerkotliana*, non-random (preferential or positive assortative) mating was found in one cross i.e. BHU × Baripada. In this cross, homogamic matings were significantly more frequent than heterogamic ones indicating evidence for preferential mating i.e. females of one strains prefer its own males. Thus these two strains show incipient sexual isolation between them which might have originated due to genetic differences resulting in the development of premating isolation. In *D. parabipectinata*, 4 strains were tested and 6 crosses were made. In three crosses e.g. Mauritius × Indonesia, Cambodia × Mysore and Mysore × Indonesia, homogamic matings were more frequent than heterogamic ones and deviation from randomness was significant. Further, isolation estimate was very low in these crosses. This clearly indicates that these strains of *D. parabipectinata* have developed mating isolation as a result of which there is preference for homogamic matings. Since these strains were derived from different ecogeographic populations, sexual isolation among these strains might have developed as a byproduct of genetic

divergence resulting from the action of natural selection and genetic drift operating in these populations. Further, the operation of genetic drift in laboratory populations may also produce genetically based divergence in mating behaviour causing ethological isolation¹⁰.

In several species of *Drosophila* such as *D. paulistorum*¹¹, *D. athabasca*¹², *D. birchii*¹³, *D. prosaltans*¹⁴, *D. silvestris*¹⁵, *D. immigrans*¹⁶, *D. ananassae*^{10,17} and *D. bipectinata* (Singh and Chatterjee, unpublished), significant deviation from random mating in interstrain crosses indicating incipient sexual isolation has been reported. Thus different populations of the same species may develop reproductive isolation as a result of genetic divergence which is an important step leading to the creation of new species. However, in certain species there is lack of evidence for divergence in mating behaviour in spite of genetic divergence in natural populations. For example, in *D. melanogaster*¹⁸ and *D. pseudoobscura*¹⁹ widely separated populations do not show mating isolation in spite of considerable amount of genetic divergence in these populations which suggests that there is considerable stability in mate recognition system of these species. However, the results obtained in *D. bipectinata*, *D. parabipectinata* and *D. malerkotliana* clearly suggest that positive assortative mating may evolve as a byproduct of genetic divergence.

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